

A major problem faced in the workshop as an Engineering Student was the need for multiple tools. From driving different nails to pulling them back, and from heavy duty applications to sensitive ones such as working with marble, different hammers are required for different tasks.

So just to solve this problem I took it on as my semester-end project — a Multifunctional Hammer.

To avoid costly mistakes in production, I first built the complete model in SolidWorks, then started manufacturing it in the workshop.

SolidWorks Modeling:

- Handle — Boss Extrude, Threads
- Head — Boss Extrude, Threads, Holes
- Attachments — Boss Extrude, Threads

Workshop Manufacturing:

- Handle — Lathe, Metric Die
- Head — Lathe, Radial Drill, Metric Tap
- Attachments — Lathe, Metric Die

From design to production, it taught me how much a good plan matters before you touch a single machine.

Skills Applied: SolidWorks Modeling · Boss Extrude · Thread Design · Lathe Operation · Radial Drilling · Tap & Die Threading · Product Design